

Microprocessor Systems

Lab11 - Introduction of I2S & MAX98357

Prof. Ren-Shuo Liu

NTHU EE



Outline

- Music Synthesis
- Introduction of I2S & MAX98357
- Example: Mario Bros
- Appendix 1:UART
- Appendix 2: General purpose IO (GPIO)

Music Synthesis

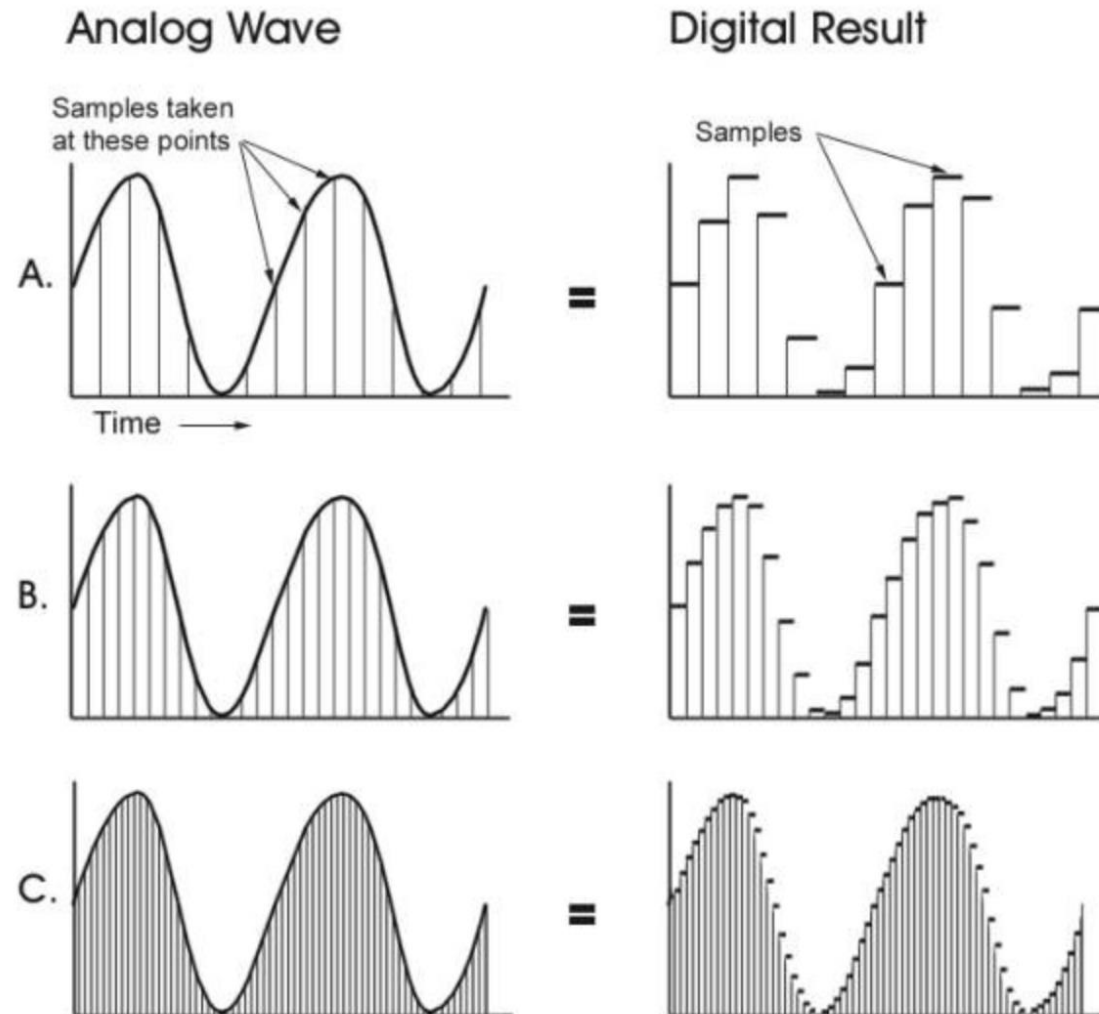
- Sampling (from continuous to discrete)
- Sound synthesis

Sampling – From Continuous to Discrete

- When you perform a digital recording, you have to convert continuous analog signal into discrete digital signal.
- To achieve this, we have to sample the signal. This simply means taking a finite number of samples of the original signal and converting it into an integer to be stored in memory.

Sample Rates

Increasing Sample Rates



Sample Rate

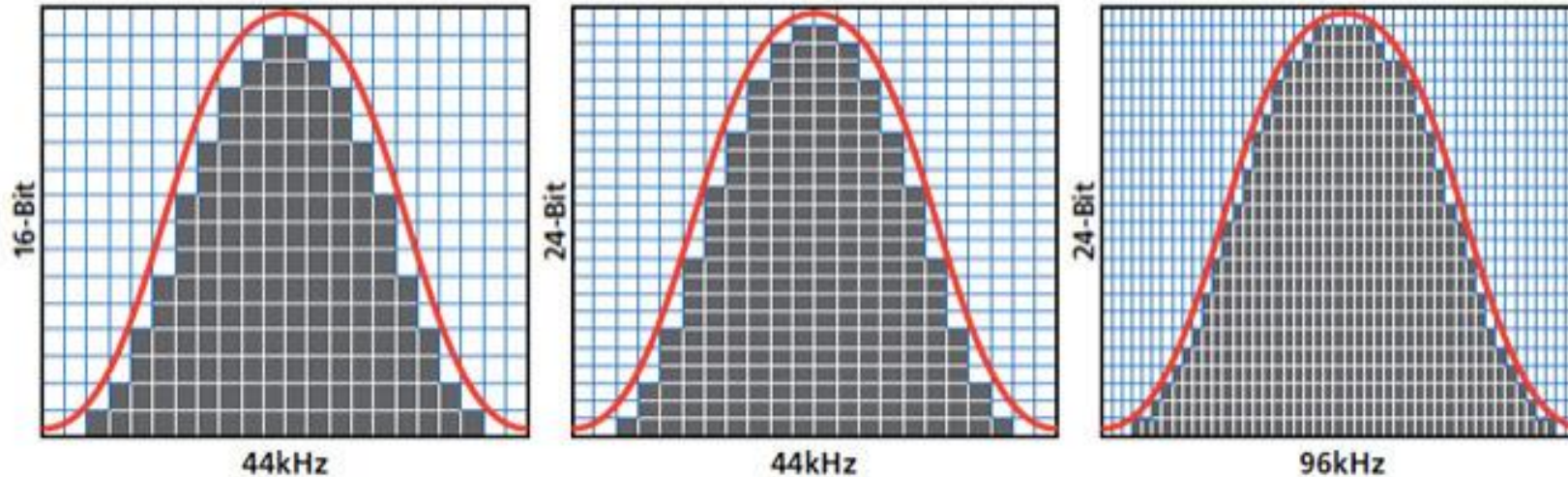
- For example, a CD audio source with a 44.1 kHz sample rate will deliver two 16-bit words 44,100 times every second.
☐ 1,411,200 bits per second ($44100 \times 16 \times 2$).
- P.S. As the human ear can not perceive audio signals above 22 kHz, if you want to exactly reproduce all the frequencies below 22 kHz, you have to use a sampling rate two times greater, which yields 44 kHz.

Sample Rate

Table 6.2: Data rate and bandwidth in sample audio applications

Quality	Sample Rate (KHz)	Bits per Sample	Mono/ Stereo	Data Rate (uncompressed) (kB/sec)	Frequency Band (KHz)
Telephone	8	8	Mono	8	0.200-3.4
AM Radio	11.025	8	Mono	11.0	0.1-5.5
FM Radio	22.05	16	Stereo	88.2	0.02-11
CD	44.1	16	Stereo	176.4	0.005-20
DAT	48	16	Stereo	192.0	0.005-20
DVD Audio	192 (max)	24 (max)	6 channels	1,200.0 (max)	0-96 (max)

Bit Depth



- If voltages are distributed between 0 to 1, but we have only 8 bits to represent them, then effectively we force all continuous values of voltage into only 256 different values.
- That is, with higher bit depth, we can get a smoother tone.

Bit Depth

- Bit depth is a matter of the audible range we can hear. The human ears can perceive sound amplitude between 0 dB SPL (sound pressure level) and 130 dB SPL.
- If a piece of music is between 10 and 100 dB SPL, it yields a dynamic range of 90 dB. A bit depth of 16 yields:

$$20 \times \log(2^{16}-1)=90.3 \text{ dB (good enough)}$$

Bit Depth

- Increase dynamic range beyond 16 bits is analogous to adding wider range to a radar used to measure the speed of cars;
- ☐ A radar that can measure speeds from 0.01 to 100,000 kilometers per hour is no more useful in this application than one that can measure speeds from 1 to 1,000 kilometers per hour.

Higher Sampling Rates' or Bit Depths' Effect

- There is one application that can benefit from higher bit depth: in professional audio recording and editing.

☐ During recording, audio often goes through many generations of processing, editing, and post-production, and each step could add noises.

☐ By using 24-bit processing, added noise could be minimized. When the audio production is finished, it can then be down-converted to 16 bits for distribution.

A Simple Example of Music Synthesizing: Sampling

$\text{♩} = 180$

G C F C G

180 beats per minute = 3 beats per second

E5 = 659.26 Hz, lasts for $1/6 = 0.16667$ seconds

D3 = 146.83 Hz, lasts for $1/6 = 0.16667$ seconds

A Simple Example of Music Synthesizing: Sampling

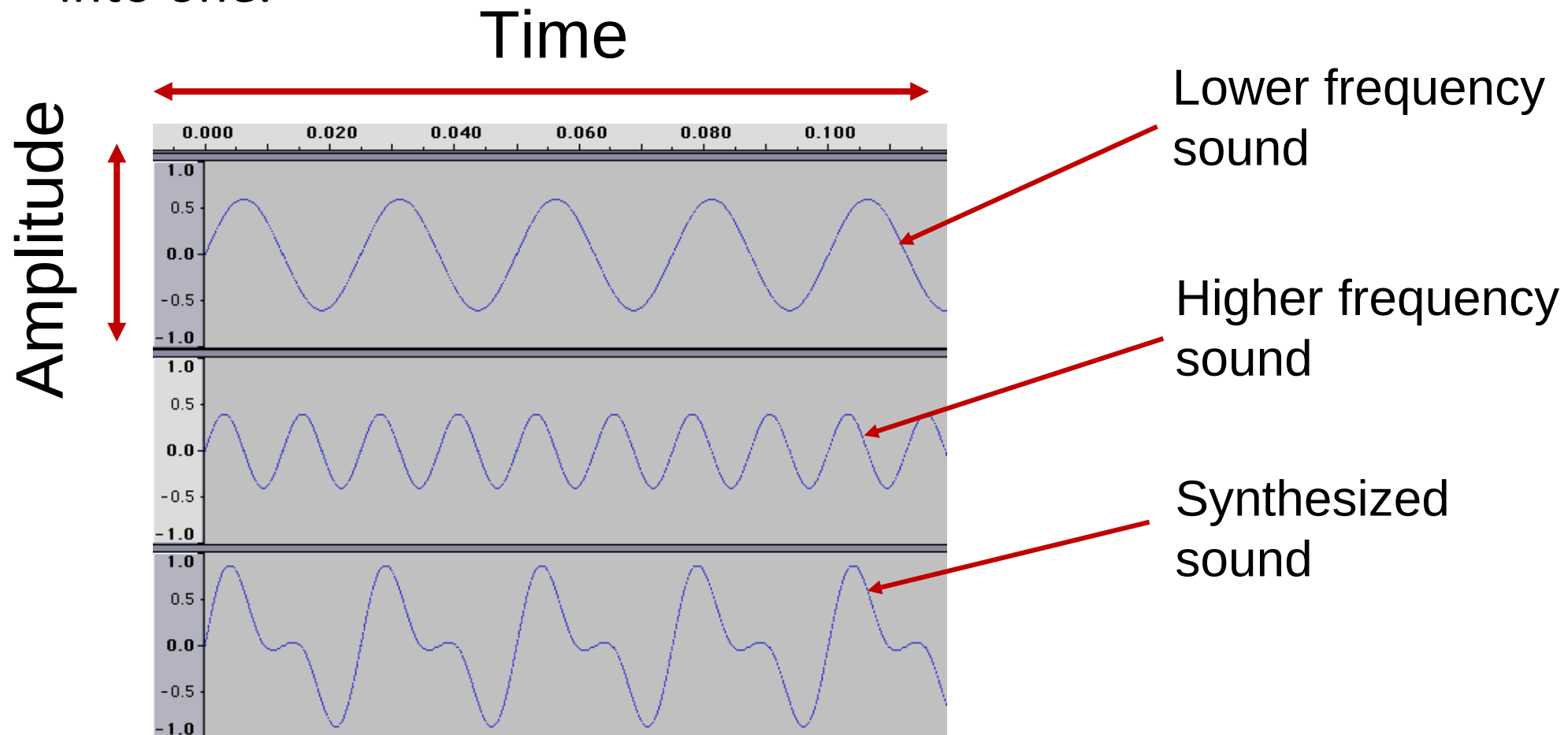


We can sample two notes (音符) to get their samples under 8 kHz sampling rate individually.

0.16667second = 1333 sample points under a sampling rate of 8kHz

After Sampling: Synthesizing

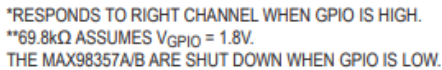
- After sampling, we could synthesize several individually sampled notes into one.



Introduction of I2S & MAX98357

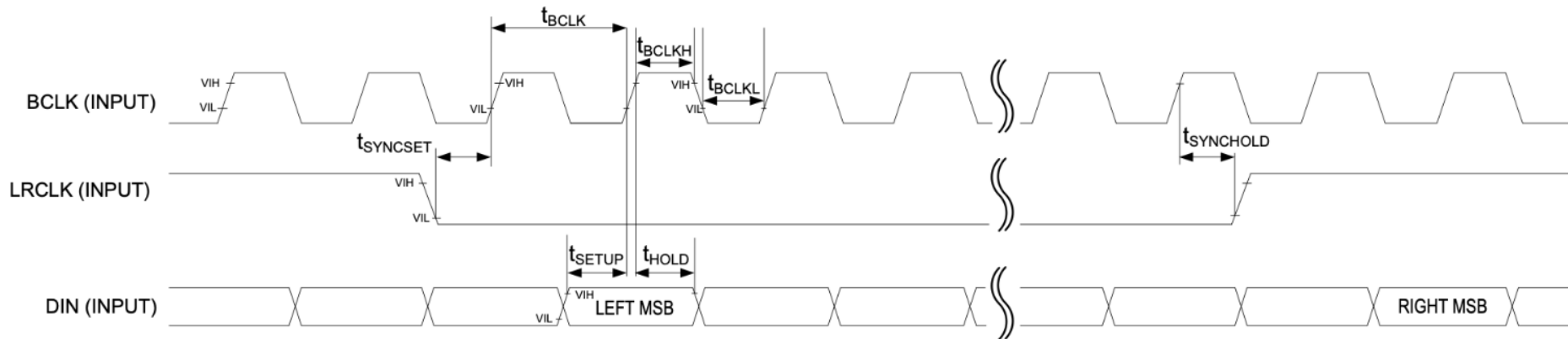
Introduction

- I²S (Inter-IC Sound), is an electrical serial bus interface standard used to connect digital audio devices together. It is used to communicate PCM (pulse code modulation) audio data between integrated circuits in an electronic device.
- This standard was introduced in 1986 by Philips Semiconductor and was last revised on June 5, 1996.

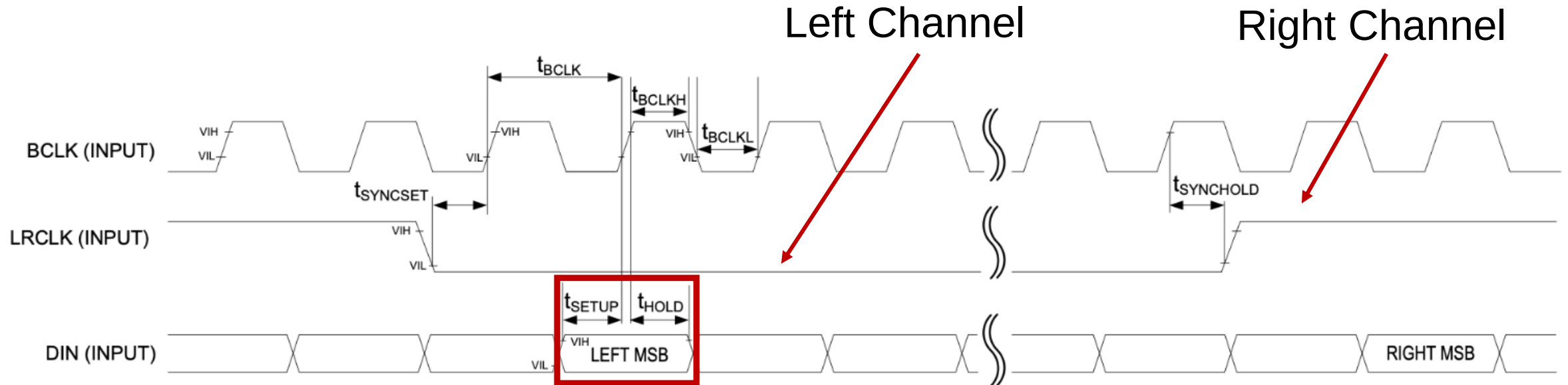


I2S Bus

- Bit clock line
 - Officially called "continuous serial clock (SCK)". Typically written as "bit clock (BCLK)".
- Word clock line
 - Officially "word select (WS)".
 - 0 = Left channel, 1 = Right channel
- Data line
 - Officially "serial data (SD) or DIN"

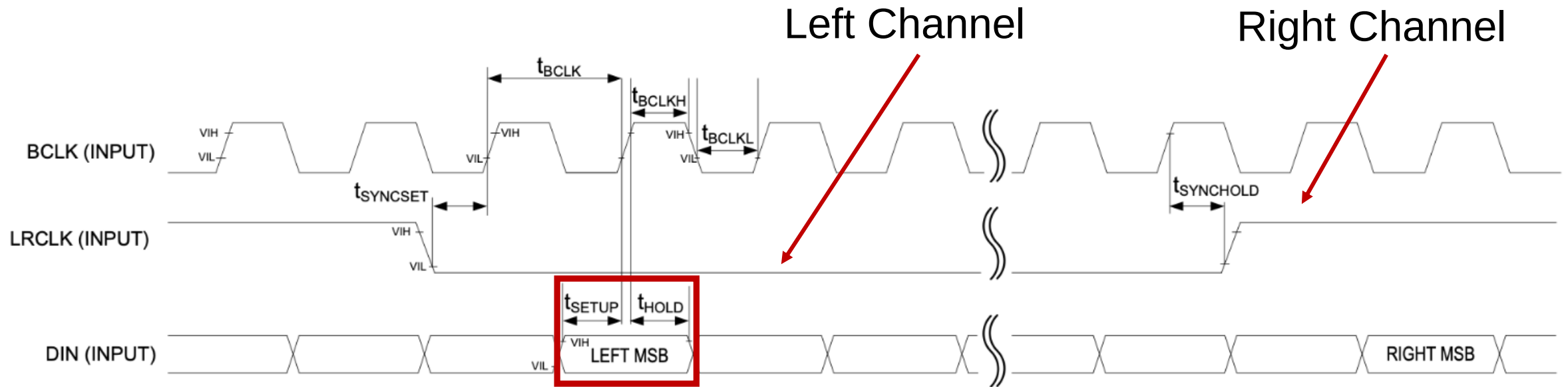


I2S Timing Diagram



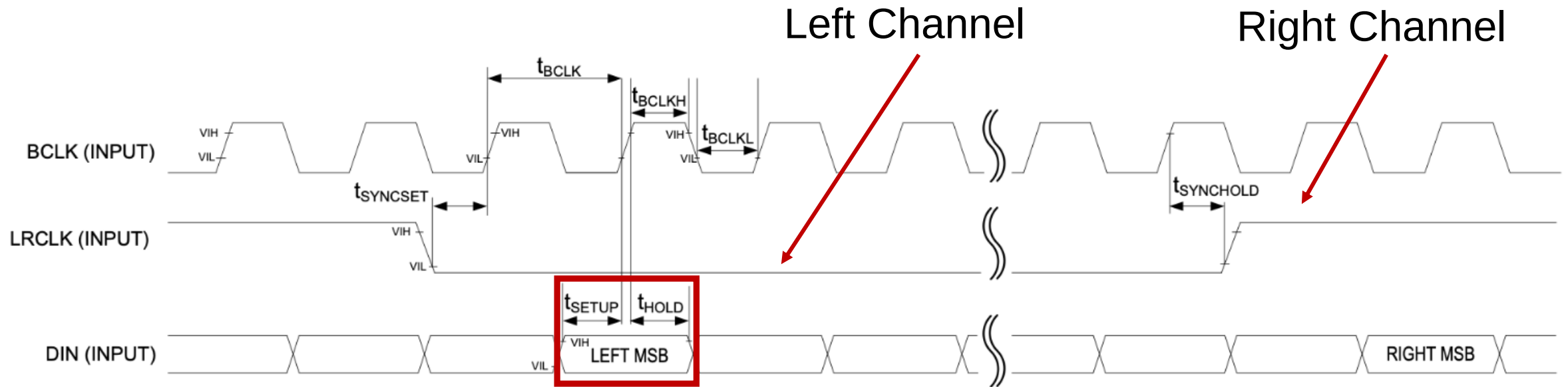
- The MSB has a fixed position, whereas the position of the LSB depends on the word length. The transmitter always sends the MSB of the next word one clock period after the WS changes.

I2S Timing Diagram



- Serial data is transmitted in 2's complement with the MSB first. The MSB is transmitted first because the transmitter and receiver may have different word lengths.

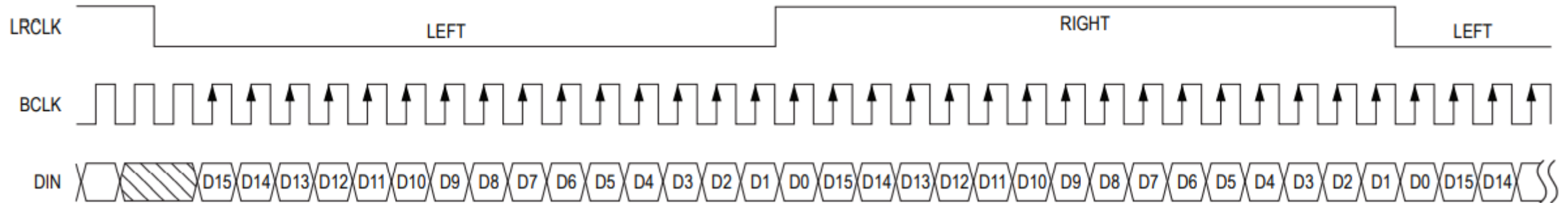
I2S Timing Diagram



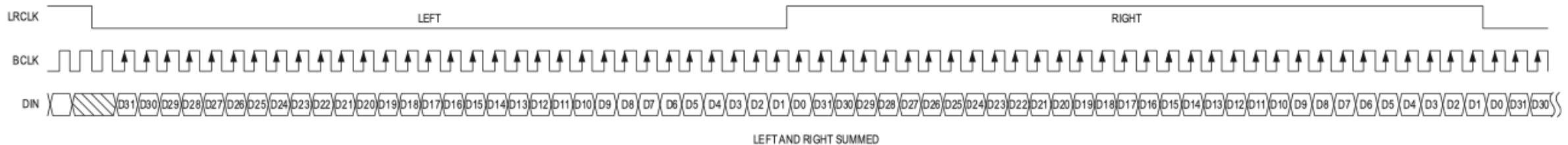
- It is unnecessary for the transmitter to know how many bits the receiver can handle, nor does the receiver need to know how many bits are being transmitted.

Overall Timing Diagram

- 16 bits/channel



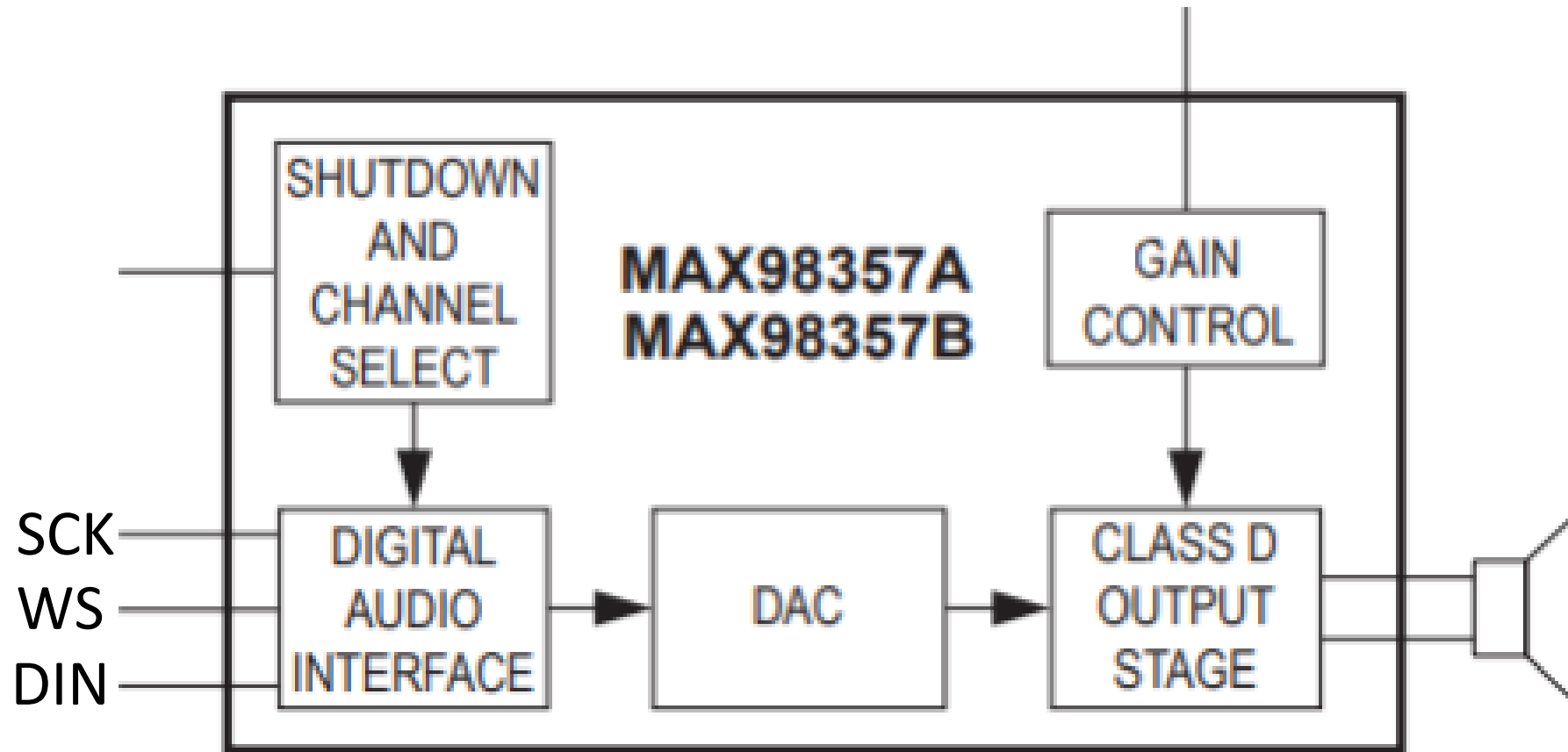
- Another example: 32 bits/channel



I2S – Length of Transmitted Data

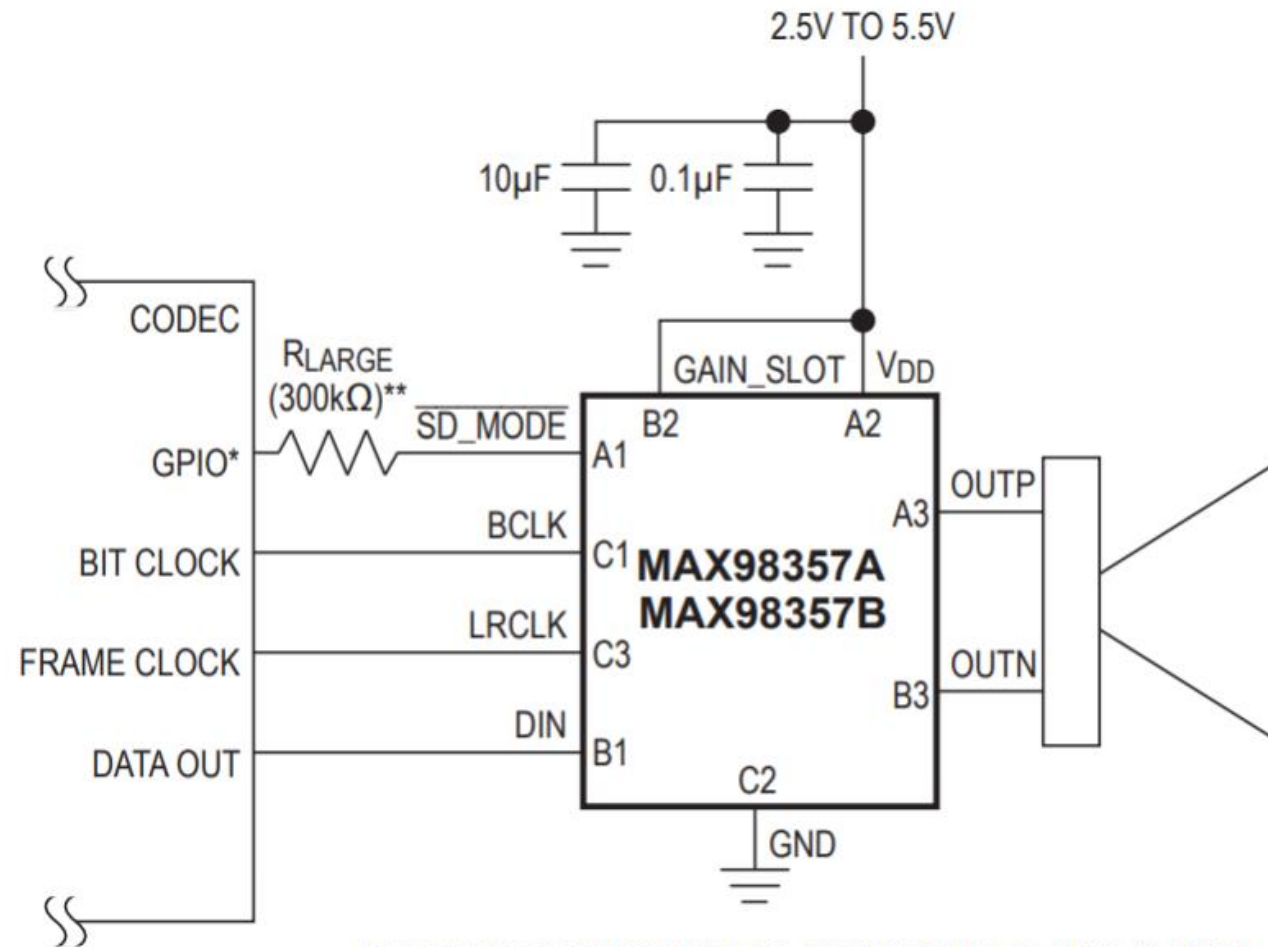
- When the system word length is greater than the transmitter word length, the word is truncated (least significant data bits are set to '0') for data transmission.
- If the receiver is sent more bits than its word length, the bits after the LSB are ignored. On the other hand, if the receiver is sent fewer bits than its word length, the missing bits are set to zero internally.

MAX 98357 Block Diagram



Drive SD_MODE low to put the ICs into shutdown.

MAX 98357 Block Diagram



*LEFT AND RIGHT CHANNELS SUMMED WHEN GPIO IS HIGH.

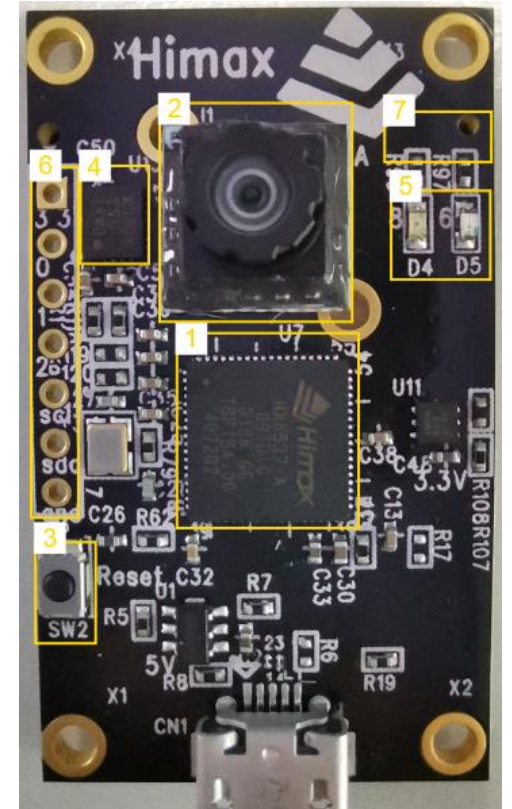
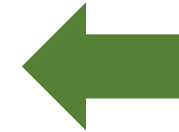
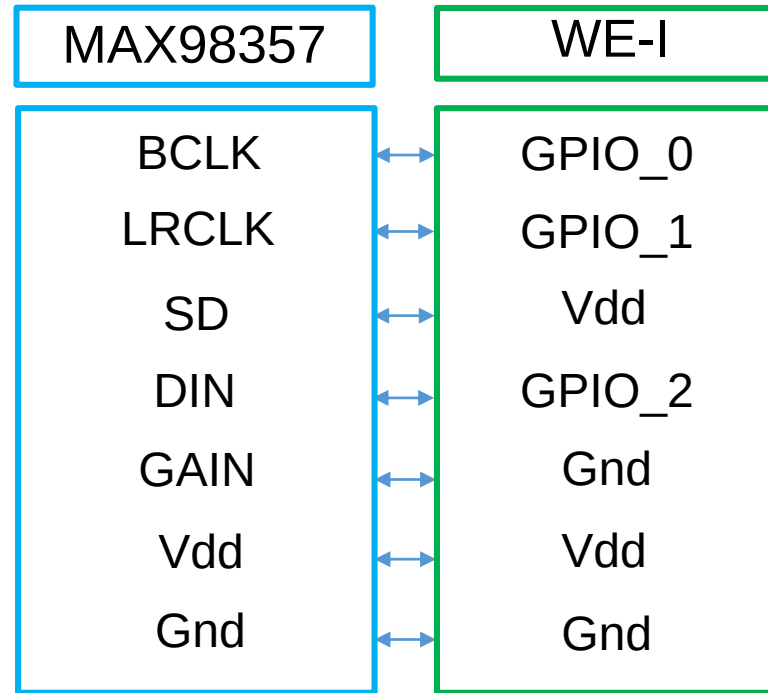
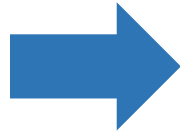
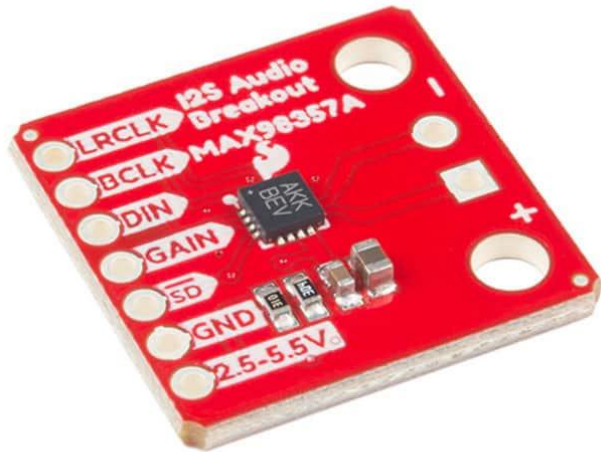
**300kΩ ASSUMES $V_{GPIO} = 1.8V$.

THE MAX98357A/B ARE SHUT DOWN WHEN GPIO IS LOW.

MAX 98357

- The input stage of the digital audio interface is highly flexible, supporting 8kHz–96kHz sampling rates with 16/24/32-bit resolution for I2S/left justified data as well as up to an 8-slot, 16-bit, or 32-bit time division multiplexed (TDM) format.

Connection



Example: Mario Bros



Appendix 1: UART

User Guide

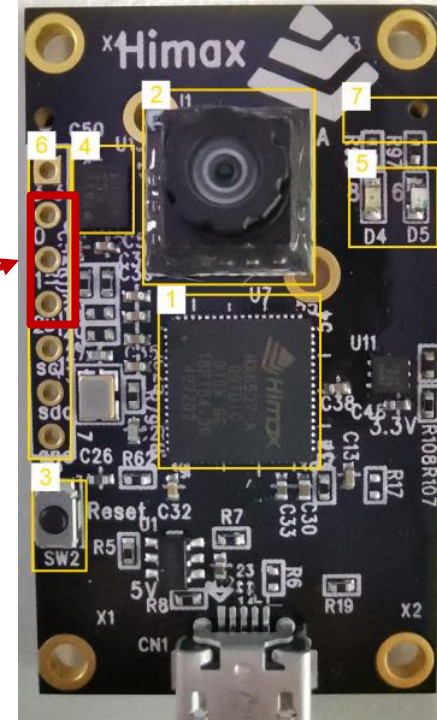
- Step 1: initialize UART (specify the baud rate)
`hx_drv_uart_initial(UART_BR_115200);`
- Step 2: send and get UART data
- Send data: `hx_drv_uart_print(const char*fmt, ...);`
- Get data: `hx_drv_uart_getchar(uint8_t *pch);`
- **NOTE:** you could print something on TeraTerm through UART to debug your code!!!

Appendix 2: General Purpose IO (GPIO)

User Guide

- A general-purpose input/output (GPIO) is a digital signal pin on an integrated circuit that may be used as an input or output, or both, and is controllable by the user at runtime.
- GPIOs have no pre-defined purpose and are unused by default, just like P0~P3 of 8051.

WE-I has 3 GPIOs



Using GPIOs

- Step 1: Declare GPIOs
Which GPIO/GPIOs are you using?
- Step 2: Set directions of GPIOs (input or output)
Use it as input or output?
- Step 3: Send initial values to GPIOs.
- Step 4: send data to GPIOs used as outputs or get data from GPIOs used as inputs.

Declare GPIO Variables

```
hx_drv_gpio_config_t hal_gpio_0;  
hx_drv_gpio_config_t hal_gpio_1;  
hx_drv_gpio_config_t hal_gpio_2;
```

Initialize GPIOs

- Which GPIO are you using? ☐ GPIO_0
- Use it as input or output? ☐ Input
- Initial value of GPIO ☐ 0 (reset)

```
hal_gpio_init(&hal_gpio_0, HX_DRV_PGPI0_0, HX_DRV_GPIO_INPUT, GPIO_PIN_RESET
```

The declared GPIO variable

Which GPIO pin?

Input or Output?

Initial Value

Send Data/Get Data

- Send data:

```
hal_gpio_set(&hal_gpio_0, GPIO_PIN_SET);
```

- Get data:

```
hal_gpio_get(&hal_gpio_2, &gpio2_data);
```

Reference

- <https://datasheets.maximintegrated.com/en/ds/MAX98357A-MAX98357B.pdf>
- <https://hackaday.com/2019/04/18/all-you-need-to-know-about-i2s/>
- <https://www.applied-acoustics.com/techtalk/sampleratebitdepth/>
- <https://zh.wikipedia.org/wiki/I%C2%B2S>
- [Synthesis Basics \(beausievers.com\)](https://beausievers.com/synthesis-basics/)
- [Music and Computers \(columbia.edu\)](https://columbia.edu/music-computers/)
- <https://ysolife.com/recording-sampling-rate/>